

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1938

Dooville 230 kV
August 01, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1938. The Interconnection Customer proposes a 50.3 MW battery energy storage facility interconnecting at the Dooville 230 kV substation in the Indiana area. The request was received on November 25, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 528 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$280,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
LONETREE 500/230 kV XFMR	EAST METHUN - LONETEE 500kV	101.5	1,081	8.3
MURDOCK - SABANERA 230kV L	E METHUEN - HENRY COUNTY 500kV	100.0	1,527	7.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LONETREE 500/230 kV XFMR	Replace transformer with higher-rated unit	\$25,024,000	1% (\$254,000)
SABANERA SUBSTATION - MURDOCK 230kV Line	Reconductor line	\$8,553,000	0% (\$26,000)
	Total allocated to project		\$280,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.